WEEK 2 · DAY 3 · QUIZ 1

Quiz 1: Weeks 1–2

30 questions · 20 minutes · just you

30 questions · 20 minutes · just you. Phones face down, notebooks closed, no AI tools. Circle **one** letter per question. True/False questions have two options.

Name: _____ Unit: _____

Structure of time-series data

1. Which property most clearly separates a time series from an ordinary table of records?

- **A.** It was exported from a database
- **B.** It has more than one numeric column
- **C.** The order of the rows carries information
- **D.** Every row has a unique ID

2. Ka-band availability drops every August during the rainy season. Which component is this?

- **A.** Noise
- **B.** Trend
- **C.** Cycle
- **D.** Seasonality

3. NOC tickets moving from 38 last week to 41 this week is a change the shift lead should investigate.

- **A.** True
- **B.** False

4. Capacity demand was strong for two years, weak for three, and is now rising again. Which component best describes this?

- **A.** A cycle, because the length is not fixed in advance
- **B.** Noise, because it is irregular
- **C.** Trend, because it is rising now
- **D.** Seasonality, because it repeats

5. Why do we call .asfreq("D") after setting a daily DatetimeIndex?

- **A.** To sort the rows by date
- **B.** To make missing days appear as NaN instead of silently disappearing
- **C.** To average duplicate readings
- **D.** To convert the timestamps to UTC

6. Telemetry is logged in UTC and the ticket system in WAT. What is the correct approach before joining them?

- **A.** Shift the tickets back by one day
- **B.** Ignore it, since the offset is only one hour
- **C.** Drop the time part and join on date only

- **D.** Convert both to the same time zone first

Visualising time-series and data storytelling

7. When is a tight y-axis range (for example 97 to 100) the right choice?

- **A.** For bounded measures that live near the top, with the range stated on the chart
- **B.** Only for bar charts
- **C.** Never, the axis must always start at zero
- **D.** For quantities you can add up, such as tickets

8. To answer "did the link breach its SLA this month?", which daily summary of hourly link margin should you plot?

- **A.** The daily mean
- **B.** The daily minimum
- **C.** The daily median
- **D.** The monthly mean

9. What is the recommended way to compare weekly uptime across six regions?

- **A.** Small multiples on a shared y-axis
- **B.** Two y-axes with three regions on each
- **C.** Six lines on one axis in different colours
- **D.** A pie chart of average uptime

10. Dividing each region's uptime by its week-1 value and multiplying by 100 lets you answer which question?

- **A.** Which region has the most terminals
- **B.** Which week had the most rain
- **C.** Which region fell furthest from its own starting point
- **D.** Which region has the highest uptime

11. A report-ready chart title should state the finding, not just the variable name.

- **A.** True
- **B.** False

12. In the four-sentence finding, which sentence comes last?

- **A.** Recommendation
- **B.** Observation
- **C.** Implication
- **D.** Situation

Rolling windows and smoothing

13. Why is a seven-day window the natural first choice for daily NOC tickets?

- **A.** It is the shortest window allowed
- **B.** It contains every weekday exactly once, so the weekend effect cancels
- **C.** It is the pandas default

- **D.** Tickets are only counted weekly

14. A trailing 30-day rolling mean is safe for live monitoring. What is its main cost?

- **A.** It removes the trend
- **B.** It cannot handle missing days
- **C.** It reacts late, roughly half a window behind the event
- **D.** It uses future data

15. A centred rolling window is a good input for a forecast because it lines up exactly with events.

- **A.** True
- **B.** False

16. s.rolling(7).mean() returns NaN for the first six days. What does that mean?

- **A.** The data has gaps
- **B.** A seven-day mean genuinely does not exist until day seven
- **C.** The series needs sorting
- **D.** The function failed

17. In .ewm(alpha=...).mean(), what does a high alpha do?

- **A.** Follows the data closely and reacts fast
- **B.** Smooths heavily and reacts slowly
- **C.** Removes the weekly cycle
- **D.** Ignores the most recent day

18. Why should SLA breaches never be reported from a smoothed line?

- **A.** Smoothed lines are harder to plot
- **B.** Smoothing adds noise
- **C.** Smoothing changes the units
- **D.** Short outages vanish inside the average

Decomposing an operational time series

19. In additive decomposition, how does the observed series relate to its parts?

- **A.** observed = trend + seasonal + residual
- **B.** observed = trend x seasonal x residual
- **C.** observed = trend - seasonal
- **D.** observed = seasonal / trend

20. Which period should you pass when decomposing daily NOC tickets?

- **A.** 365
- **B.** 12
- **C.** 24
- **D.** 7

21. The seasonal swing grows as the level rises. Which model fits?

- **A.** Neither, use a straight line
- **B.** Either, they give the same result

- **C.** Multiplicative
- **D.** Additive

22. Which component should the NOC incident review look at for genuinely unusual months?

- **A.** The trend
- **B.** The raw series
- **C.** The residual
- **D.** The seasonal profile

23. How many full seasonal cycles does the course recommend at minimum before trusting a seasonal profile?

- **A.** Ten
- **B.** One
- **C.** Two
- **D.** Three

24. One catastrophic August makes every August look bad in the seasonal profile. What is the suggested fix?

- **A.** Use a multiplicative model
- **B.** Use STL with robust=True
- **C.** Switch to period=7
- **D.** Delete that August

Autocorrelation in operational data

25. What is autocorrelation at lag 1?

- **A.** The average change per day
- **B.** The correlation of a series with a copy of itself shifted by one step
- **C.** The correlation between two different series
- **D.** The slope of the trend

26. A correlogram of daily throughput shows spikes at lags 7, 14 and 21. What does this tell you?

- **A.** The data is daily noise
- **B.** The series has no memory
- **C.** There is a weekly season with period 7
- **D.** There is a strong trend

27. A correlogram where every lag stays high and decays slowly usually indicates what?

- **A.** Pure noise
- **B.** A random split
- **C.** A trend
- **D.** A backlog cleared in cycles

28. The bar at lag 0 of a correlogram is always 1.0 and can be ignored.

- **A.** True
- **B.** False

29. Why is a random train/test split invalid on autocorrelated data?

- **A.** scikit-learn does not allow it
- **B.** Neighbouring rows are near-copies, so the test set leaks into training
- **C.** It is slower to compute
- **D.** It removes the seasonal profile

30. Suya sales and generator sales both rise steadily over time and correlate at 0.97, but their changes (.diff()) correlate at only 0.06. What explains the raw correlation?

- **A.** A real link between the two businesses
- **B.** The weekly roster
- **C.** Measurement error
- **D.** Mostly shared trend